

XENIX[®] System V/286 Operating System

Operations Guide

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Chapter 1

Introduction

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1.1 Overview

The XENIX[®] operating system is a powerful system of programs that helps you perform a full spectrum of tasks, from developing high-level and assembly language programs to creating, editing, and typesetting documents. To keep this powerful system running smoothly, the XENIX system requires careful control of its operation and a regular schedule of maintenance. This guide explains how to operate and maintain the XENIX operating system on your computer, ensuring maximum performance with the least number of system problems.

This guide also explains how to expand a XENIX system with a Micnet network. A Micnet network allows serial communication between all the XENIX systems in your work environment. The Micnet programs and commands include the **netutil** program, which is used to install the network, and the **mail**, **rcp**, and **remote** commands, which are used to pass messages, files, and commands over the network.

1.2 The System Manager

Every XENIX system should have at least one person in charge of system maintenance and operation. In this guide, that person is called the system manager. It is the system manager's duty to ensure the smooth operation of the system and to perform tasks that require special privileges.

Depending on the size of the system and the number of users on the system, a system manager's job can be anything from a once-a-week task to a full-time job. Even if the system is small, the system manager should faithfully perform each required maintenance task, since sloppy maintenance can affect XENIX performance.

All tasks in this guide are presented from the system manager's point of view, but many can also be accomplished by ordinary users. Since some of the tasks dramatically change the system's operation, we recommend that, whenever possible, the system manager perform these tasks. This can prevent unwanted or unnecessary changes to the system.

1.3 The Super-User Account

The super-user account is a special account for performing system maintenance tasks. It gives the system manager unusual privileges that ordinary users do not have, such as accessing all files in the system and executing privileged commands. Many of the tasks presented in this guide require that the system manager be logged in as the super-user. To do this, the system manager must know the super-user password created during the installation of the XENIX system (see the *XENIX Installation Guide*).

Users who are authorized to act as the super-user, including the system

manager, should log in as the super-user only when it is necessary to perform a system maintenance task. The system manager should create a personal user account and use it for day-to-day work, reserving the super-user account for system maintenance tasks only, even if the system manager is the only person using the system.

The number of individuals who are given the super-user password should be kept to a minimum. Misuse of the super-user powers by naive users can result in a loss of data, programs, and even the XENIX system itself.

1.4 The Keyboard

Many keys and key combinations have special meanings in the XENIX system. These keys and key combinations have special names that are unique to the XENIX system and may or may not correspond to the keytop labels on your keyboard. To help you find the special keys, the following table shows which keys on a typical console correspond to XENIX system keys. In this table, a hyphen (-) between keys means you should hold down the first key while pressing the second.

XENIX Name	Keytop	Action
BREAK	DELETE	Stops current action and returns to the shell. This key is also called the INTERRUPT key.
BACKSPACE	BACKSPACE	Deletes the first character to the left of the cursor.
CTRL-D	CTRL-D	Signals the end of input from the keyboard; also exits current shell.
CTRL-H	ERASE	Deletes the first character to the left of the cursor. Also called the ERASE key.
CTRL-Q	CTRL-Q	Restarts printing after it has been stopped with CTRL-S.

CTRL-S	CTRL-S	Stops printing at the standard output device (does not stop the program).
CTRL-U	CTRL-U	Deletes all characters on the current line. Also called the KILL key.
CTRL-\	CTRL-\	Quits current command and creates a <i>core</i> file (Recommended for debugging only).
ESCAPE	ESC	Exits the current mode; for example, exits insert mode when in the editor <i>vi</i> .
RETURN	RETURN	Terminates a command line and initiates an action from the shell.

1.5 Using This Guide

The tasks presented in this guide range from simple tasks requiring little knowledge about XENIX, to complex tasks requiring extensive knowledge about XENIX and your computer.

Each chapter explains the tools and knowledge you need to complete the tasks described in that chapter. In some cases, you may need to refer to other manuals, such as the *XENIX User's Guide*.

Chapter 1 introduces this guide.

Chapter 2 explains how to start and stop the XENIX system and how to log in as the super-user, the XENIX system's special system manager account.

Chapter 3 explains how to create accounts for the users who work on your system, how to assign groups, and how to manage user IDs.

Chapter 4 explains how to create and mount file systems, how to set permissions, and how to keep the system secure.

Chapter 5 explains how to maintain free space on the root file system and oth-

er file systems.

Chapter 6 explains how to create backup copies of the root file system and other file systems.

Chapter 7 explains how to add terminals and printers to the system.

Chapter 8 explains how to solve system problems such as a jammed line-printer or a forgotten password.

Chapter 9 explains how to create a multiple system mailing network with Micnet.

Chapter 10 explains how to install device drivers.

Appendix A presents a list of the XENIX system special files and explains how to use these files when creating and maintaining file systems.

Appendix B presents a list of commonly used XENIX directories and log files.

Appendix C explains how to build a communication link to other XENIX systems.

Chapter 2

Starting and Stopping the System

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2.1 Introduction

This chapter explains how to start and stop the XENIX system. It also explains how to log in as the super-user.

2.2 Starting the System

Starting a XENIX system requires more than just turning on the power. You must also perform a series of steps to initialize the system for operation. Starting the system requires:

- Loading the operating system
- Cleaning the file system (if the system was improperly stopped)
- Choosing the mode of system operation

The following sections describe each of these procedures.

2.2.1 Loading the Operating System

The first step in starting the system is to load the operating system from the computer's hard disk. Follow these steps:

1. Turn on power to the computer and hard disk. The computer loads the XENIX bootstrap program and displays the boot prompt:

```

SYSV      XENIX Boot
Enter:    hd program
          fd program
          dos
          cf [-c conf_file] [device program]

Press Enter for default:    hd /xenix
:
```

2. Press the RETURN key. The bootstrap program loads the XENIX operating system.

When the system is loaded, it displays information about itself and checks to see if the "root file system" (i.e., all files and directories) is clean. If it is clean, you may choose the mode of operation. If not, the system requires you to clean the file system before choosing.

2.2.2 Cleaning the File System

You must clean the file system if the system displays the message:

```
Proceed with cleaning (y or n)?
```

This message indicates that the system was not stopped properly as described later in this chapter in Section 2.4. The XENIX operating system requires a clean file system to perform its tasks.

To clean the file system, type `y` (for yes) and press the RETURN key. The system cleans the file system, repairing damaged files or deleting files that cannot be repaired. It reports on its progress as each step is completed. At some point, it may ask if you wish to salvage a file. Always answer by typing `y` and pressing the RETURN key.

The option to not clean the file system or salvage a file is available for diagnostic purposes only.

When cleaning is complete, the system usually asks you to choose the mode of operation, but it may also display the message:

```
** Normal System Shutdown **
```

If it displays this message, you must reload the system. You can do this by resetting the computer and repeating the steps given in the previous section. For instruction on how to reset your computer, see the hardware manual provided with the computer.

2.2.3 Choosing the Mode of System Operation

You may choose the mode of XENIX operation as soon as you see the message:

```
Type Ctrl-d to continue with normal startup,  
(or give the root password for system maintenance):
```

The system has two modes: normal operation and system maintenance mode. Normal operation is for ordinary work on the system. This is the mode you should choose to allow multiple users to log in and begin work. System maintenance mode is a specialized mode reserved for work the system manager must do. It does not allow multiple users.

To choose normal operation, press the CTRL-D key sequence. The system displays a startup message and begins to execute commands found in the command file `/etc/rc` described in Chapter 8, "Solving System Problems." When the commands have finished, the system displays the "login:" prompt. You may then log in as a normal user, as described in the *XENIX User's Guide*, or

as the super-user, as described in Section 2.3.

To choose system maintenance mode, type the super-user's password (sometimes called the "root password") and press the RETURN key. The system displays the message of the day and the maintenance mode prompt (#). The commands in the */etc/rc* file are not executed. (Choose system maintenance mode only if you must do system maintenance work that requires all other users to be off the system.)

2.3 Logging In As the Super-User

Many system maintenance tasks, when performed during normal operation, require that you log in as the super-user. For example, you must be logged in as the super-user to stop the system.

Before you may log in as the super-user, you need the super-user password. You also need to see the "login:" message on your terminal's screen. If you do not see this message, press the CTRL-D key until it appears.

To log in as the super-user, follow these steps:

1. When you see the "login:" message, type the super-user's login name:

root

and press the RETURN key. The system asks for the super-user's password.

2. Type the super-user's password and press the RETURN key. The system does not display the password as you type it, so type each letter carefully.

The system opens the super-user account and displays the message of the day and the super-user prompt (#).

Take reasonable care when you are logged in as the super-user. In particular, you should be very careful when deleting or modifying files or directories. Avoid using wildcard designators in filenames and frequently check your current working directory. Small errors can cause annoying and unwanted changes to the system and user files. Some errors can cause irretrievable damage to a file or the system.

You can leave the super-user account at any time by pressing CTRL-D.

2.4 Stopping the System

Stopping the XENIX system takes more than just turning off the computer. You must prepare the system for stopping by using either the **shutdown** or the

haltsys command. The following sections describe each command.

2.4.1 Using the shutdown Command

The **shutdown** command is the normal way to stop the system and should be used whenever the system is in normal operation mode. It warns other users that the system is about to be stopped and gives them an opportunity to finish their work.

To stop the system with the **shutdown** command, follow these steps:

1. Log in as the super-user (see Section 2.3). The system opens the super-user account and displays the message of the day and the super-user's prompt.
2. Type:

```
/etc/shutdown
```

and press the RETURN key. The system loads the command, which, in turn, asks for the number of minutes you wish to elapse before the computer stops:

```
Minutes till shutdown? (0-15):
```

3. Type any number from 0 to 15 and press the RETURN key. The system displays a warning message at each terminal, asking logged in users to finish their work and log out. As soon as all users are logged out or the specified time has elapsed, the system closes all accounts, displays the message:

```
** Normal System Shutdown **
```

and stops.

You may now turn off the computer.

2.4.2 Using the haltsys Command

The **haltsys** command may be used to halt the system immediately. In general, it should be used only when no other users are on the system or when the system is in system maintenance mode.

To stop the system with the **haltsys** command, follow these steps:

1. Log in as the super-user (not required when in system maintenance mode). The system opens the super-user account and displays the message of the day and the super-user prompt.

2. Type:

```
sync ; /etc/haltsys
```

and press the RETURN key. The system displays the message:

```
** Normal System Shutdown **
```

and stops. It is important to use the **sync** command so that the file system will be clean the next time XENIX is reloaded.

You may now turn off the computer.





Chapter 3

Preparing XENIX for Users

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3.1 Introduction

User accounts help the XENIX system manager keep track of the people using the system and control their access to the system's resources. Ideally, each user should have a user account. Each account has a unique *login name* and *password* with which the user enters the system, and a *home directory* where the user works.

It is the system manager's job to create accounts for all users on the system. It is also the manager's job to maintain user accounts by changing user passwords, login groups, and user IDs when necessary.

This chapter explains how to:

- Add user accounts to the system
- Change an account's password
- Force new passwords
- Create a group
- Change an account's login group
- Change an account's user ID
- Remove user accounts from the system

The following sections describe each task in detail.

3.2 Adding a User Account

You may add a user account to the system with the **mkuser** program. The program creates a new entry in the XENIX system's */etc/passwd* file. This entry contains information about the new user, such as login name and initial password, that the system uses to let the user log in and begin work. The program also creates a home directory for the user, a mailbox for use with the **mail** command, and a *.profile* file, which contains XENIX commands that are executed when the user logs in.

To create a new user account, follow these steps:

1. Log in as the super-user.
2. Type:

mkuser

and press the RETURN key. The system displays the following message:

```
Newuser
-----
Add a user to the system
```

Do you require detailed instructions? (y|n|q):

3. Type the letter *y* (for “yes”) if you want information about the program, otherwise, type the letter *n* (for “no”). (Type *q* (for “quit”) only if you wish to stop the program and return to the system.)

When the program continues, it asks you to enter the new user’s login name:

Enter new user’s login name:

The login name is the name by which XENIX will know the user. It is usually a short version of the user’s actual name, typed in lowercase letters. For example, either “*johnd*” (a first name and last initial) or “*jdoe*” (a first initial and last name) is acceptable for the user John Doe.

4. Type the new name and press the RETURN key. The program asks you if you want to use the default group:

Do you want to use the default group? (y|n|q):

The group name is the name of the group of users to which the new user will belong when the user logs in. Users in a group have access to a common set of files and directories. The XENIX system’s default group is “*group*” (with group ID 50). If you want to use this group, type *y* and press RETURN. If you want another group, type *n* and follow the self-explanatory instructions.

5. Next, the program asks you for the initial password:

Enter password:

The initial password is the password you assign to the new user. The user will use the initial password to enter the account for the first time. Once in the account, the user should create a new password, one that is hard to guess. (See the *XENIX User’s Guide* for more information about changing passwords.)

6. Type the password carefully and press the RETURN key. After you have entered the password, the program asks you to re-enter it again. This is

a safety check on the spelling of the password. If the two passwords do not match, you will be asked to try again. Next, the program prompts you for a shell type:

ENTER Shell type (1,2,3 or 4) and press ENTER:

7. The shells available are: 1) Standard (Bourne) Shell, 2) Visual Shell, 3) Cshell, and 4) /usr/lib/uucp/uucico. Enter the number that corresponds to the shell you want to use and press the RETURN key. After you have entered the shell number, the program asks for a comment:

```
Please enter Comment >-----
>
```

A comment is information about the new user, such as department name and phone extension. Although the comment is optional, it is useful if you often use the **finger** command to display information about users. If given, the comment must be no more than 20 characters long, including spaces. It must not contain any colons (:). The example:

John Doe, 123

shows the recommended form for a comment.

8. Type the comment. Make sure it is 20 characters or less. If you do not wish to enter a comment, just press the RETURN key.

The program now shows the information you have entered and creates an entry for the user that will be copied to the special system file */etc/passwd*. The display shows the login name and user ID, the group name and number, the comment field, and the shell.

9. The program then gives you an opportunity to change the user name, password, group, or comment:

Do you want to change anything? (y|n|q):

10. Type the letter *y* (for "yes") and press the RETURN key, if you wish to change something. Type *n* (for "no") and skip to step 12 if you wish to complete the new account. (Type *q* (for "quit") only if you wish to leave the program and cancel the new account.)
11. If you type *y* to make changes, the program asks for the item you wish to change:

username

```
group
password
comment
shell
```

Type the name of the item you wish to change, then type the new item. The program changes the item and redisplay the information on the user.

12. Once you type *n* for no changes, the program displays the message:

```
Password file updated
```

followed by a description of the actions it has taken to add the new user account to the system. The program then asks if you wish to add another user to the system.

13. Type *y* if you wish to add another user. Otherwise, type *n* to stop the program and return to the super-user prompt.

A user can log into a new account as soon as it is created. See the *XENIX User's Guide* for details.

3.3 Changing a User's Password

Normally, ordinary users can change the passwords of their own accounts with the **passwd** command (see the *XENIX User's Guide*). Sometimes, however, it may be necessary for the super-user to change the password, for example, if the user has forgotten the password and cannot get into the account to change it. The super-user may change the password of any user (including the super-user) with the **passwd** command.

To change a password, follow these steps:

1. Log in as the super-user.
2. Type:

```
passwd login-name
```

(where *login-name* is the user's login name) and press the RETURN key. The command displays the message:

```
New password:
```

3. Type the new password and press the RETURN key. The command does not display the password as you type it, so type carefully. The com-

mand asks you to type the password again:

Retype new password:

4. Type the password again and press the RETURN key.

To see how ordinary users can change their own passwords with the **passwd** command, see the *XENIX User's Guide*.

3.4 Forcing a New Password

From time to time, a user account may need a higher level of security than ordinary. Since the security of any account depends on its password, it is important to keep the password as secret as possible. One way to provide greater security is to force users to change their passwords on a regular basis.

You can force users to change their passwords by using the **pwadmin** command. This command automatically dates each password and requires the user to provide a new password when the specified number of weeks have passed. The command also requires users to wait a minimum number of weeks before allowing them to restore their previous password. To use the **pwadmin** command, you must log in as the super-user.

You can enable password aging for a specified user by using the **-a** option. Type:

```
pwadmin -a login-name
```

where *login-name* is the login name of a user. The user will then be required to wait a minimum number of weeks before the password can be changed, and will be forced to change the password after a maximum number of weeks have elapsed. The **-a** option uses the default minimum and maximum values found in the */etc/default/password* file.

You can choose your own minimum and maximum number of weeks by using the **-min** and **-max** options. For example, a common pair of minimum and maximum values is 2 and 8. To set the minimum and maximum dates, type:

```
pwadmin -min num -max num login-name
```

where *num* is a number in the range 0 to 63, and *login-name* is simply the login name of the user whose password you are administering. Note that the minimum and maximum cannot both be 0, and that the minimum must not be greater than the maximum.

If you are unsure of the current minimum and maximum values for a password, you can display them by typing:

```
pwadmin -d login-name
```

This command does not change the current values.

If you wish to force a user to change a password immediately, type:

```
pwadmin -f login-name
```

The user is asked on the next login to supply a new password.

When a password no longer requires extra security, you can remove the current minimum and maximum values for the password by typing:

```
pwadmin -n login-name
```

The system will no longer prompt for changes.

For more information about password aging, see *pwadmin(C)* and *passwd(M)* in the *XENIX Reference Manual*.

3.5 Creating a Group

A group is a collection of users who share a common set of files and directories. The advantage of groups is that users who have a common interest in certain files and directories can share these files and directories without revealing them to others. Initially, all users belong to the common system group named “group”, but you can create new groups by modifying the XENIX system file */etc/group* using a XENIX text editor.

To create a new group, you need to choose a group name and a group identification number (group ID). You also need to make a list of the users in the new group. The group name may be any sequence of letters and numbers up to eight characters long, and the group ID may be any number in the range 0 to 65535. Both the group name and ID must be unique, i.e., they must be not be the same as any existing group name or ID.

To create a new group, follow these steps:

1. Log in as the super-user.
2. Display the contents of the */etc/group* file by typing:

```
cat /etc/group
```

then press the RETURN key. The **cat** command displays the contents of the */etc/group* file. The file contains several entries, each defining the group name, group ID, and users for a group. Each entry has the form:

group-name::group-ID:users

The users are shown as a list of login names separated by commas (.). For example, a typical file may look like this:

```
other:x:1:demo
sys:x:2:
group::50:johnd,suex
```

3. Check the */etc/group* file entries to see that the group name and ID you have chosen are unique.
4. If the group name and ID are unique, invoke a XENIX text editor (see the *XENIX User's Guide*) and specify */etc/group* as the file to edit.
5. Locate the last line in the file, then insert the new entry in the form given above. For example, if you wish to create a group named "shipping" with group ID "142" and users "johnd", "marym", and "suex", type:

```
shipping::142:johnd,marym,suex
```

6. Exit the editor.

To make sure you have entered the group names correctly, use the **grpcheck** command to check each entry in the */etc/group* file. If the new entry is free of errors, no other changes to the file are required.

You can create any number of new groups. Each group may have any number of members. Furthermore, any user may be a member of any number of groups. Multiple group membership is especially convenient for users who have interests that span a variety of areas.

If users are members of several groups, they can gain access to each group by using the **newgrp** command. See the *XENIX User's Guide* for details.

3.6 Changing a User's Login Group

When users log in, the system automatically places them in their "login group". This is the group given by the group ID in the user's */etc/passwd* file entry (see Section 3.2 in this chapter). You can change the user's login group by changing the group ID. To change the group ID you need the group ID of the new login group, and you need to know how to use a XENIX text editor (see the *XENIX User's Guide*).

To change the group ID, follow these steps:

1. Log in as the super-user.
2. Use the **cd** command to change the current directory to the */etc* directory. Type:

```
cd /etc
```

3. Use the **cp** command to make a copy of the */etc/passwd* file. Type:

```
cp passwd passwd+
```

4. Invoke a text editor and specify */etc/passwd+* as the file to edit.
5. Locate the desired user's password entry. Each entry begins with the user's login name.
6. Locate the user's group ID number in the user's password entry. It is the fourth item in the entry. Items are separated by colons (:). For example, the entry:

```
marym:9iKlwp:205:50:Mary March, 122:/usr/marym:/bin/sh  
has group ID "50".
```

7. Delete the old group ID and insert the new one. Be sure you do not delete any other portion of the user's password entry.
8. Exit the editor.
9. Use the **mv** command to save the old */etc/passwd* file. Type:

```
mv passwd passwd-
```

10. Use the **mv** command to make the edited file the new */etc/password* file. Type:

```
mv passwd+ passwd
```

You can make sure you have entered the new login group correctly by using the **pwcheck** command. If the new entry is correct, no other changes to the file are required.

You must not change the group IDs for system accounts such as "cron" and "root". System accounts are any accounts whose user IDs are less than 200. The user ID is the third item in the password entry.

Note that changing a user's login group does not change the "group ownership" of the user's files. Group ownership defines which group has access to

a user's files. If users in the new group wish to access the user's files, you must change the group ownership with the **chgrp** (for "change group") command. For details, see Section 4.4.2 in Chapter 4, "Using File Systems."

3.7 Changing a User ID

Sometimes it is necessary to change the user ID in a user's account entry to allow a user to access files and directories transferred from other computers. In particular, if a user has different accounts on different computers and frequently transfers files and directories from one computer to another, then the user IDs in each of the user's account entries must be made the same. You can make them the same by modifying the account entries in the */etc/passwd* file.

To change a user ID, follow these steps at every computer for which the user has an account:

1. Log in as the super-user.
2. Use the **cd** command to change the current directory to the */etc* directory. Type:

```
cd /etc
```

3. Use the **cp** command to make a copy of the */etc/passwd* file. Type:

```
cp passwd passwd+
```

4. Invoke a XENIX text editor and specify */etc/passwd+* as the file to edit.
5. Locate the user's account entry. Each entry begins with the user's login name.
6. Locate and substitute the current user ID. The ID is the third item in the entry. For example, the entry:

```
marym:9iKlwp:205:50:Mary March, 122:/usr/marym:/bin/sh
has user ID "205".
```

7. Exit the text editor.
8. Use the **mv** command to save the old */etc/passwd* file. Type:

```
mv passwd passwd-
```

9. Use the **mv** command to make the edited file the new */etc/passwd* file. Type:

```
mv passwd+ passwd
```

No other changes to the file are required.

In most cases, you can change the user ID to the same number as the user's most-used account. But the new number must be unique at every system for which the user has an account. If there is any conflict (for example, if the number already belongs to another user on one of the systems), you must choose a new number. You can choose any number greater than 200. Just make sure it is unique, and that you copy it to all systems on which the user has an account.

Once a user's ID has been changed, you must change the "user ownership" of the user's files and directories from the old user ID to the new one. You can do this with the **chown** (for "change owner") command described in Chapter 4, "Using File Systems." For example, to change the ownership of johnd's home directory, type:

```
chown johnd /usr/johnd
```

Note that you may use the **find** command described in Chapter 5, "Maintaining File Systems," to locate all files and directories with the user's old user ID.

3.8 Removing a User Account

It is sometimes necessary to remove a user account from the system. You can remove a user account with the **rmuser** program. The program deletes the user's entry from the */etc/passwd* file and removes the user's home directory and mailbox.

Before you can remove the user account, you must remove all files and directories from the user's home directory or move them to other directories. If you wish to save the files, you may use the **tar** command to copy the files to a floppy disk (see Section 6.4.1 in Chapter 6, "Backing Up File Systems").

To remove a user account, follow these steps:

1. Log in as the super-user.
2. Type:

```
cd /usr/login-name
```

and press the RETURN key to change to the user's home directory. The *login-name* must be the user's login name.

3. Make sure that you have made copies of all important files and direc-

tories in the user's home directory.

4. Use the **rm** (for "remove") command to remove all files and directories from the user's home directory. This includes any files that begin with a period (.). Directories can be removed by using the **-r** (for "recursive") option of the **rm** command. For example, the command:

```
rm -r bin
```

removes the directory named *bin* and all files within this directory.

5. After removing all files and directories, make sure the user's mailbox is empty. Type:

```
cat /usr/spool/mail/login-name
```

and press the RETURN key. The *login-name* is the user's login name. If the mailbox contains text, type:

```
cat /dev/null >/usr/spool/mail/login-name
```

and press the RETURN key.

6. When the user's home directory and mailbox are empty, type:

```
cd /usr
```

and press the RETURN key. The user's home directory cannot be removed until you have moved to another directory.

7. Type:

```
rmuser
```

and press the RETURN key. The program displays a message explaining how to remove a user:

```
****rmuser-remove a user from the system****
```

```
Press ENTER when you are ready.
```

The program asks for the login name of the user you wish to remove:

```
Enter name of id to be removed.
```

8. Type the user's login name. You should now see the message:

```
Removing user name from the system.  CONFIRM? (y/n/q):
```

where *name* is the login name of the user you want to remove.

9. Type *y* (for “yes”) to remove the user from the system. Otherwise, type *n* (for “no”) to stop the removal, or *q* (for “quit”) to stop the program. The program removes the user’s entry from the */etc/passwd* file, the user’s mailbox, *.profile* file, and home directory. The program displays the message:

```
User name removed from the system
```

where *name* is the login name of the user you have removed.

The program now gives you a chance to remove another user:

```
Do you want to remove another user? (y/n/q) :
```

10. Type *y* to remove another user. Otherwise, type *n* or *q* to stop the program.

Note that the **rmuser** program will refuse to remove an account that has a system name, such as “root”, “sys”, “sysinfo”, “cron”, or “uucp”, or a system ID (user ID below 200). Also, the program cannot remove a user account if the user’s mailbox still has mail in it, or if the user’s home directory contains files other than *.profile*.

Chapter 4

Using File Systems

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4.1 Introduction

This chapter describes one of the most important responsibilities of a system manager: controlling and recording users' access to the files and directories on the system. It introduces file systems, permissions, system security, and process accounting.

4.2 File Systems

A file system is the XENIX system's way of organizing files on mass storage devices such as hard and floppy disks. A file system consists of files, directories, and the information needed to locate and access these items.

Each XENIX system has at least one file system. This file system is called the root file system and is represented by the symbol `"/"`. The root file system contains all the XENIX programs that may be used by the system manager. It usually contains all the user directories as well.

A XENIX system may also have other file systems that contain user directories and application programs. One reason for using other file systems is to expand the available storage space of the system. Each additional file system adds its free space to the system's total space. New file systems can be specifically created by a user, then mounted onto the system so they can be used.

You can create a file system with the **mkfs** command. This command sets the size and format of the file system and may also copy some files to the new system. You can mount a file system with the **mount** command. When you have mounted the file system, you may access the files and directories in the file system in the same way as files and directories in the root file system. (The root file system is permanently mounted.) When you are finished with a file system, you can unmount it with the **umount** command.

You can create file systems on hard and on floppy disks. A reason for creating new file systems on floppy disks is to establish a collection of application programs and data files that can be easily mounted and used when needed.

The following sections explain how to create and use file systems.

4.2.1 Creating a File System

You can create a file system on a floppy disk by using the **makefs** command. To create the file system, you need:

- A floppy disk
- The special filename of a floppy disk drive

Note that if a file system already exists on the disk, it will be destroyed by this procedure. For this reason, be particularly careful not to create a new file system on the root file system. If you destroy the root file system, you will have to reinstall the XENIX system.

To make a file system on a floppy disk, follow these steps:

1. Log in. You do not have to be logged in as the super-user to use the **makefs** command.
2. Insert a floppy disk into a floppy disk drive. Make sure there is no read-only tab on the disk jacket.
3. Type:

```
makefs
```

Follow the instructions on the screen. For more information on the **makefs** command, type:

```
help makefs
```

4.2.2 Mounting a File System

Once you have created a file system, you can mount it with the **mount** command. To mount a file system you need:

- The special filename of a disk drive
- The name of an empty directory

The special filenames of disk drives are given in Appendix A, “XENIX Special Device Files.” The directory to receive the file system may be any directory that is empty (contains no files) and is not your current working directory. Note that the directory */mnt* is specifically reserved for mounted file systems.

To mount a file system, follow these steps:

1. Log in. You do not have to be logged in as the super-user to use this command.
2. Insert the disk that contains the file system into a floppy disk drive.
3. Type the appropriate **mount** command and press the RETURN key. The command should have the form:

```
/etc/mount specialfile directoryname
```

where *specialfile* is the special filename of the disk drive containing the disk and *directoryname* is the name of the directory to receive the file system. If the disk has a read-only tab, make sure you include the switch **-r** at the end of the command.

For example, you may use the following command to mount the disk in disk drive */dev/fd1* onto the directory named */account*:

```
/etc/mount /dev/fd1 /account
```

Remember to make sure that the specified directory is empty before issuing the command. If the command displays the message:

```
mount: Structure needs cleaning
```

use the **fsck** command to clean the file system and try to mount it again (see Section 5.3 in Chapter 5, "Maintaining File Systems"). If the command displays the message:

```
mount: Device busy
```

either the file system has already been mounted and cannot be mounted twice, or a user is currently in the directory. You must wait for users to leave a directory before you can mount the directory.

To check that the file system was properly mounted, use the **cd** command to change to the directory containing the mounted system and the **ls** command to list the contents. The command displays the files and directories in the file system. Be sure to use the **cd** command to leave the directory after finishing your work in it.

Note that frequently used file systems can be mounted automatically when starting the system by appending the appropriate **mount** commands to the */etc/rc* file. See Section 8.11.1 in Chapter 8, "Solving System Problems" for details.

4.2.3 Unmounting a File System

You can unmount a mounted file system with the **umount** command. Unmounting a file system does not destroy its contents. It merely removes access to the files and directories in the file system.

To unmount a mounted file system, type:

```
/etc/umount specialfile
```

Press the RETURN key. The *specialfile* is the name of the special file that corresponds to the disk drive that contains the disk with the file system. The command empties the directory that previously contained the file system and makes the directory and the corresponding disk drive available for mounting another file system.

For example, the following command unmounts a file system from the disk drive */dev/fd1*:

```
/etc/umount /dev/fd1
```

Before unmounting a file system, make sure that no files or directories are being accessed by other commands or programs. The **umount** command displays the message:

```
umount: Device busy
```

if you or another user is currently in the directory containing the file system.

4.2.4 Formatting Floppy Disks

You can format floppy disks with the **format** program. Formatted disks are required whenever you create a file system. They are also required when you back up a file system with the **sysadmin** program (see Chapter 6, “Backing up File Systems”).

To format a floppy disk, follow these steps:

1. Insert a disk into floppy disk drive 0. Make sure there is no read-only tab on the disk jacket.
2. Type:

```
format /dev/devicename
```

Press the RETURN key. The program formats the disk.

Warning: The **format** command is machine-specific. Each OEM (original

equipment manufacturer) must provide its own software and documentation for the format command.

If for some reason you have not properly inserted the disk into the drive, the program displays an error message and stops.

In general, the system manager should format spare floppy disks in advance. Note that formatting removes all data from the disk, so if you are formatting a disk that already contains data, make sure that the data is nothing you wish to save.

4.3 Permissions

Permissions control access to all the files and directories in a XENIX system. In XENIX, ordinary users may access those files and directories for which they have permission. All other files and directories are inaccessible.

There are three different levels of permissions: user, group, and other. User permissions apply to the owner of the file; group permissions apply to users who have the same group ID as the owner; and other permissions apply to all other users.

4.3.1 Displaying Permissions

You can display the permission settings for all the files in a directory with the `ls -l` (for “list directory”) command. This command lists the permissions along with the name of the file’s owner, the size (in bytes), and the date and time the file was last changed. The command display has the following format:

```
-rw-rw---- 1 johnd group 11515 Nov 17 14:21 file1
```

The permissions are shown as a sequence of ten characters at the beginning of the display. The sequence is divided into four fields. The first field (the “type” field) has a single character, the other fields (“user”, “group”, and “other”) have three characters each. The characters in the fields have the following meanings.

In the type field:

- d Indicates the item is a directory
- Indicates the item is an ordinary file
- b Indicates the item is a device special block I/O file
- c Indicates the item is a device special character I/O file

In the user, group, and other fields:

- r Indicates read permission. Read permission for a file means you may copy or display the file. Read permission for a directory means you may display the files in that directory.
- w Indicates write permission. Write permission for a file means you may change or modify the file. Write permission for a directory means you may create files or subdirectories within that directory.
- x Indicates execute permission (for ordinary files) or search permission (for directories). Execute permission for a file means you may invoke the file as you would a program. Execute permission for a directory means you may enter that directory with the `cd` command.
- Indicates no permission.

For example, the permissions:

```
-rwxrwxrwx
```

indicate an ordinary file with full read, write, and execute access for everyone (user, group, and other).

The permissions:

```
-rw-----
```

indicate an ordinary file with read and write access for the user only.

The permissions:

```
drwxr-x--x
```

indicate a directory with search access for everyone, read access for the user and group, and write access for only the user.

When you create a file, the XENIX system automatically assigns the following permissions:

```
-rw-r--r--
```

This means that everyone may read the file but only the user may write to it. When you create a directory, the system assigns the permissions:

```
drwxr-xr-x
```

field in */etc/ttys*.

- initial-flags* Set the terminal line characteristics when **getty** first establishes the connection with the line. **Getty** recognizes the flags listed in *tty(M)* in the *XENIX Reference Manual*. Often, the only flag that appears in this field is the one that sets the baud rate. For example, B300 sets the speed to 300.
- final-flags* Set the terminal line characteristics just before **getty** executes the **login** command. These flags describe the operating characteristics for the line. The baud flag (B) sets the baud rate again. Other common flags include SANE (a composite flag that sets a number of terminal characteristics to reasonable values), TAB3 (expands tabs with spaces), IXANY (enables any character to restart output), and HUPCL (hangs up line on final close). You can enter flags in any order.
- login-prompt* Contains the login prompt message that greets users. The login message will be displayed exactly as you type it in this field, including spaces and tabs.
- next-label* Identifies the next label in *gettydefs* for **getty** to try if the current one is not successful. **Getty** tries the next label if a user types the BREAK key while attempting to log in to the system. Groups of entries, for example, for dial-up lines or for TTY lines, should form a closed set so that **getty** cycles back to the original entry if none of the entries is successful.

Each field is separated by pound sign (#), and each entry in *gettydefs* is separated by a blank line.

An entry in *gettydefs* might look like this:

```
4# B1200 # B1200 SANE TAB3 HUPCL #Login: #2
```

The number 4 identifies this entry to **getty**. The next field sets the baud rate to 1200. The final settings in the third field include the baud rate (B1200), SANE (a composite flag for a number of characteristics), and HUPCL (hangs up line on final close). The login prompt appears as "Login:", and if this setting is not successful, **getty** proceeds to label 2 in *gettydefs*.

7.3.2 Changing the Gettydefs File

The file */etc/gettydefs* already exists in your XENIX system, and has sets of entries for the operator's console, dial-up lines, and terminal lines. These dif-

ferent sets correspond to line mode settings in the */etc/ttys* file. When the **init** program initializes the XENIX system, it passes line mode information to **getty**.

You can edit *gettydefs* to add new terminal settings or to change existing ones. For example, the settings for terminal lines on your XENIX system might look like this:

```
1# B2400 # B2400 SANE IXANY TAB3 #Login: #2
2# B4800 # B4800 SANE IXANY TAB3 #Login: #3
3# B9600 # B9600 SANE IXANY TAB3 #Login: #1
```

To change the sample *gettydefs* file so that the first baud rate **getty** attempts to set is 1200, do the following:

1. Log in as super-user.
2. Enter a text editor to edit the first line of the file *gettydefs*.
3. Change B2400 in both the second and third fields to B1200.
4. Exit the text editor, saving *gettydefs*.

The sample file should look like this:

```
1# B1200 # B1200 SANE IXANY TAB3 #Login: #2
2# B4800 # B4800 SANE IXANY TAB3 #Login: #3
3# B9600 # B9600 SANE IXANY TAB3 #Login: #1
```

You can also add terminal line settings to *gettydefs*. Flags and permissible values for terminal settings are listed in *tty(M)* in the *XENIX Reference Manual*.

When you add a new entry, be sure that the groups of entries in *gettydefs* form a closed set, so that the *next-label* field of the last entry directs **getty** back to the first entry in the group.

To add an entry for a baud rate of 300 to the preceding sample *gettydefs* file, follow these steps:

1. Log in as super-user.
2. Enter a text editor to edit the file */etc/gettydefs*.

3. Locate the point where you want to insert the new settings for *gettydefs*. The order of the entries does not matter; **getty** looks only for the label. In this example, the new entry will be the last entry in the file.
4. Press the RETURN key after the last line in the file and type the following on a new line:

```
4# B300 # B300 SANE IXANY TAB3 #Login: #1
```

5. To incorporate label 4 into the set of labels, change the *next label* field for entry 3 to 4:

```
3# B4800 # B4800 SANE IXANY TAB3 #Login: #4
```

Getty is now directed from label 3 to label 4, then back to label 1.

6. Exit the text editor, saving the revised *gettydefs* file.

The new *gettydefs* looks like this:

```
1# B1200 # B1200 SANE IXANY TAB3 #Login: #2
```

```
2# B4800 # B4800 SANE IXANY TAB3 #Login: #3
```

```
3# B9600 # B9600 SANE IXANY TAB3 #Login: #4
```

```
4# B300 # B300 SANE IXANY TAB3 #Login: #1
```

7.3.3 Checking the Terminal Settings

Each time you change the terminal line settings or add new entries to *gettydefs*, you should check to make sure that the new values are allowed by **getty**. To do this, use the **getty** command, the check option **-c**, and the filename.

For example, to check *gettydefs*, type:

```
getty -c /etc/gettydefs
```

If any of the values and settings in *gettydefs* are not permitted, **getty -c** will print these on your terminal screen.

For more information on **getty** and *gettydefs*, see *getty(M)* and *gettydefs(F)* in the *XENIX Reference Manual*.

7.4 Changing Serial Line Operation

Whenever you enable a terminal with the **enable** (C) command, the system automatically sets the operating characteristics of the serial line to a set of default values. Sometimes these values do not match the values the terminal uses, so you must change them to allow communication between the system and the terminal. You can display and change the operating characteristics of a serial line with the **stty** (for “set tty”) command.

You can display the current operating characteristics of a serial line by typing:

```
stty
```

at the terminal connected to that line. If it is impossible to log in at that terminal, you may use another terminal to display the characteristics. Log in as the super-user at another terminal, and type:

```
stty <specialfile
```

where *specialfile* is the name of the device special file that corresponds to the serial line (see Appendix A, “XENIX Special Device Files”). For example, the command:

```
stty </dev/tty00
```

displays the current characteristics of the serial line named */dev/tty00*. The command displays the baud rate, the parity scheme, and other information about the serial line. The meaning of this information is explained in **stty**(C) in the *XENIX Reference Manual*.

One common change to a serial line is to change the baud rate. This is usually done from a terminal connected to another serial line, since changing the rate disrupts communication between the terminal and system. Before you can change the rate, you need to know the current baud rate of the terminal (review the terminal’s hardware manual to see how to determine the terminal’s current rate). Once you have the baud rate, log in as the super-user at the other terminal, and type:

```
stty baud-rate <specialfile
```

where *baud-rate* is the terminal’s current baud rate, and *specialfile* is the name of the device special file that corresponds to the serial line you wish to change. The baud rate must be in the set 50, 75, 110, 134, 150, 200, 300, 600, 1200, 2400, 4800, 9600, exta, and extb. For example, the command:

```
stty 9600 </dev/tty00
```

changes the baud rate of the serial line */dev/tty00* to 9600. Note that the “less

than'' symbol (<) is used for both displaying and setting the serial line from another terminal.

Another common change is to change the way the system processes input and output through the serial line. Such changes are usually made from the terminal that is connected to the serial line. For example, the command:

```
stty -tabs
```

causes the system to expand tabs with spaces (used with terminals which do not expand tabs on their own), and the command:

```
stty echoe
```

causes the system to remove a character from the terminal screen when you back over it with the BACKSPACE key.

Note that the **stty** command may also be used to adapt a serial line to an unusual terminal, or to another type of serial device that requires parity generation and detection or special input and output processing.

For a full description of this command, see **stty(C)** in the *XENIX Reference Manual*.

7.5 Setting the Terminal Type

The XENIX system requires that an enabled terminal's type be clearly defined before any work is done at the terminal. You can set the terminal type by assigning the type to the **TERM** variable, a special XENIX system variable that associates the terminal you are using with a list of characteristics given in the */etc/termcap* file. The characteristics tell the system how to interpret your terminal's keys and how to display data on your terminal's screen.

The **TERM** assignment has the form:

```
TERM=termtype ; export TERM
```

The *termtype* must be one of the names associated with one of the terminals defined in the */etc/termcap* file. The assignment must be typed at the terminal whose type you are setting.

For example, to set the terminal type to "ansi", go to the terminal you wish to set, type:

```
TERM=ansi ; export TERM
```

and press the RETURN key.

If you are not sure which name you may use for *termtype*, you can view the names by displaying the */etc/termcap* file. To display the file, type:

```
cat /etc/termcap
```

and press the RETURN key. Since the file is quite large, you'll need to use the CTRL-S key sequence to stop the display at every screenful. You may view more of the file by pressing the CTRL-Q key sequence or the SPACEBAR.

You can let the system define the terminal *.profile* type automatically whenever you log in by including the TERM assignment in your *.profile* file (see Section 8.11.2 in Chapter 8, "Solving System Problems").

If you do let the system set the terminal type, be careful when you log in on terminals that are not the same as your normal terminal. The XENIX system has no way of checking whether or not the terminal assignment is correct for the given terminal, and assumes that it is the same as your normal terminal. If it is not, you must set the terminal type manually.

7.6 Removing a Terminal

From time to time it may be necessary to remove a terminal from the system, for example, if you wish to replace it with some other device. Before you can remove a terminal, you must disable it with the **disable** command.

To remove a terminal, follow these steps:

1. Turn off the power to the terminal.
2. Log in as the super-user at another terminal.
3. Use the **disable** command to disable the terminal. The command has the form:

```
disable specialfile
```

where *specialfile* is the name of the serial line to which the terminal is attached. For example, the command:

```
disable /dev/tty00
```

disables the terminal connected to serial line */dev/tty00*.

4. Disconnect the terminal from the system.

The serial line that was previously connected to the terminal is now free to accept another device.

When using the **disable** command, make sure that you wait a full minute between each use of the command. Failure to do so can cause a system crash.

7.7 Adding a Lineprinter

This section explains how to add new lineprinters to your XENIX system and how to use the lineprinter commands to organize and control your printing.

The XENIX lineprinter spooling system is a collection of commands that help you to efficiently install, monitor, and control the lineprinters serving your system. When a user requests to print a file using the **lp** command, the lineprinter system responds with a *request ID*. This consists of the name of the printer the file will be printed on and a unique number identifying the file. With this request ID, the user can find out the status of the print request or cancel it. The **lp** options help the user control printer output. For more information on **lp**, see the *XENIX User's Guide*.

A few terms need to be defined first. A *device* is the target for **lp** output. It could be a hard-wired printer, a terminal that is sometimes used as a printer, or a regular file. A device can be represented by a full XENIX pathname. A *printer* is the name assigned by the system manager to represent a device. This name can have up to 14 characters. At different times, a printer may be associated with different devices. Printers can be grouped together into *classes* which are ordered lists of printers. A *destination* can be a printer or a class.

Consult the hardware manuals provided for your computer and lineprinters for information on making the physical connection between your system and printing devices.

7.7.1 Installing a Printer: **lpinit**

To install new printing devices on your XENIX system, use the **lpinit** command. Before you use **lpinit** you should first know the port to which the lineprinter is connected or the XENIX pathname of the device (for example, */dev/lp0*) and the lineprinter interface program. A set of model interface programs is supplied with your XENIX system. See **lp(M)**. For more information on printer interface programs, see Section 7.7.9.

When you give the **lpinit** command, you will be asked a series of questions for which the default answers are displayed. If you wish to choose the default answer, simply press the RETURN key. If you want to supply your own answers, type in the information as you are prompted. If you make a mistake while responding to the questions, just press the DELETE key or the INTERRUPT key and start again.

The following example adds a lineprinter to your XENIX system. The printer name is *printer1*, the device pathname is */dev/lp0*, and the printer will be the

default printer for your system.

1. Type the command:

```
/etc/lpinit
```

2. **Lpinit** displays the following message on your screen:

The printer is attached to a

1. Serial/Parallel Adapter, Parallel Port.
2. Alternate Serial/Parallel Adapter, Parallel Port.
3. Monochrome and Printer Adapter.

Enter one of the options above

(default = 1) :

3. Select the appropriate option, type the corresponding number, and press the RETURN key.

4. **Lpinit** prompts:

```
Do you need special handling for Tandy Carriage Returns and Line Feeds ? [y|n]
[Refer to the lpinit command documentation for further information]
(default = n)
```

If your printer needs special translations (interpretive mode), type y (for “yes”) and press RETURN. If your printer runs in non-interpretive mode, simply press RETURN for the default. Most industry standard printers will use the non-interpretive mode; however, some Tandy printers will use the interpretive mode. See **lpinit(C)** and **lp(M)** for more information on interpretive and non-interpretive modes and Tandy printer handling.

5. **Lpinit** displays the following message:

```
Enter a name for the printer
(default = linepr) :
```

Printer names may be up to 14 characters long and may be any combination of numbers, letters, or underscore characters. Type the printer name, *printerI*, and press the RETURN key.

6. **lpinit** displays the following message:

```
Enter an interface program
(default = /usr/spool/lp/model/dumb)      :
```

This example uses the default choice, */usr/spool/lp/model/dumb*, so simply press the RETURN key.

7. After you have finished responding to these three questions, **lpinit** displays the following message:

```
scheduler stopped
```

This means that the lineprinter scheduling program, **lpsched**, has been momentarily stopped so that **lpinit** can add *printer1* to your system.

8. After stopping the scheduler, **lpinit** asks if this printer will be the system default printer:

```
Is this the default printer ? (y/n)
(default = y)      :
```

You can type *y* (for “yes”) or *n* (for “no”) depending upon whether you want users’ print requests to be automatically routed to *printer1* or not. Pressing the RETURN key is the same as typing *y*. Next, **lpinit** automatically gives the instructions to XENIX to enable *printer1* to accept print requests.

Note

Answering ‘yes’ to the default question provides System III compatible locking features. Use this for System III applications that do their own printing. With this setting, XENIX internally adds a prefix of an underscore (*_*) to the printer name.

After you have responded to these four questions, **lpinit** re-starts **lpsched**, and users can begin printing files on the new printer.

You can also add printers to your system using the **lpadmin** command discussed in the Section 7.7.3; however, you will need to give separate commands to stop **lpsched**, to enable the printer, and to allow it to accept print requests. For more information on these programs and commands, see Sections

7.7.2, 7.7.7, and 7.7.8.

7.7.2 Stopping the Print Spooling Daemon: **lpsched**

A *daemon* is a process that runs in the background, independent of a terminal, and performs a function. The spooling daemon, **lpsched**, routes print requests through the correct printer interface program and then to the lineprinter. You cannot print on your system unless **lpsched** is running. The **lpsched** program starts automatically each time you restart your XENIX system. Sometimes, it is necessary to stop **lpsched**, especially if you want to reconfigure printers or if you want to add new printers using the **lpadmin** command (**lpinit** automatically stops and restarts **lpsched** for you).

This section explains how to find out whether or not **lpsched** is running, how to stop and re-start it, and how to re-create it if necessary.

To find out whether or not **lpsched** is running, type:

```
lpstat -r
```

In response to the **lpstat** command, your screen displays a message that **lpsched** is either running or not running.

To shut down the scheduler, **lpsched**, type:

```
/usr/lib/lpshut
```

Lpsched stops running and all printing stops as well. Printing requests stopped in the middle of printing reprint when **lpsched** starts again.

After you have finished configuring the printers, you should restart **lpsched**. To do this, type:

```
/usr/lib/lpsched
```

Next, check **lpstat -r** to confirm that **lpsched** is running. If it is not, type the following on two lines:

```
rm -f /usr/spool/SCHEDLOCK  
/usr/lib/lpsched
```

This removes any previous version of the file, *SCHEDLOCK*. *SCHEDLOCK* assures that only one copy of **lpsched** runs at a time.

Each time **lpsched** sends a print request to an interface program, it records an entry in a log file, */usr/spool/lp/log*. The entry includes the user name, the request ID, the name of the printer the request will be printed on, and the date and time requested. **Lpsched** also records any error messages in this file.

After you have stopped **lpsched**, the log file is renamed */usr/spool/lp/oldlog* and **lpsched** starts a new log file. Requests waiting to be printed before **lpsched** was stopped may have an entry in both log files.

For more information on **lpsched**, see **lpsched(C)** in the *XENIX Reference Manual*.

7.7.3 Changing Printer Configuration: **lpadmin**

After you have added printers to your system using **lpinit**, you may want to alter their configuration. For example, you may want to modify an interface program or create a new class of printers. To change printer configuration, use the **lpadmin** command with the **-p** option. If **lpsched**, the printing scheduler, is running, **lpadmin** will not alter the configuration except where noted. To stop **lpsched**, see Section 7.7.2.

The **lpadmin** command has the form :

```
/usr/lib/lpadmin -p printer options
```

where *printer options* include the following:

- c** Creates a class of printers or adds a printer to a class. The same rules apply to class names as to printer names: up to 14 alphanumeric characters or underscores. Print requests sent to a class of printers will be printed on the first available member of the class.

For example, to add a printer named *laser* to a class of printers named *class1*, type:

```
/usr/lib/lpadmin -plaser -cclass1
```

- r** Removes *printer* from a class of printers. If the printer is the last member of the class, **lpadmin** removes the class as well. For example, to remove *laser* from *class1*, type:

```
/usr/lib/lpadmin -plaser -rclass1
```

- i** Assigns a new interface program to *printer*. The pathname of the interface program must follow **-i**. For example, to establish */usr/spool/lp/model/dumb* as the interface program for a printer named *prt1*, type:

```
/usr/lib/lpadmin -pprt1 -i/usr/spool/lp/model/dumb
```

You can also use the **lpadmin** command to add new printers to your system. Unlike the **lpinit** command, **lpadmin** only configures the new printer. You

must separately perform the steps to stop **lpsched**, enable the new printer, and give the **accept** command so that the printer will accept print requests. For more information on using **lpadmin** to add new printers to your system, see **lpadmin(C)** in the *XENIX Reference Manual*.

7.7.4 Creating a Default Printer

You can specify one printer or one class of printers on your system to be the default destination of a print request. All files will be sent to the system default destination for printing unless the user specifies a printer when using the **lp** command. For more information on **lp** and its options, see the *XENIX User's Guide*.

To create or change the system default destination, use **lpadmin** and the **-d** option. You can change this option even when the scheduler, **lpsched**, is running.

For example, to make a class of printers called *classb* the system default destination, type:

```
/usr/lib/lpadmin -dclassb
```

To establish no default destination, you use the **-d** option without a printer name:

```
/usr/lib/lpadmin -d
```

For more information, see **lpadmin(C)** in the *XENIX Reference Manual*.

7.7.5 Removing Printers

You can remove printing destinations, either printers or classes of printers, if there are no print requests routed to them. If you remove a class of printers, the individual printers remain in the system. Removal of the last remaining printer in a class of printers removes the class as well.

To remove a printer or class of printers, use **lpadmin** and the **-x** option. For example, to remove a printer, *printer2*, from your system, type:

```
/usr/lib/lpadmin -xprinter2
```

If you wish to temporarily remove a printer because of a breakdown, see Section 7.7.8. If you wish to temporarily re-route print requests to other printers, see Section 7.7.6 or Section 7.7.7.

For more information on **lpadmin** and its options, see **lpadmin(C)** in the *XENIX Reference Manual*.

7.7.6 Moving Requests Between Printers: `lpmove`

You can move print requests between printing destinations by using the `lpmove` command. This command refuses to move print requests while the scheduler, `lpsched`, is running. To stop `lpsched`, see Section 7.7.2. `lpmove` will move individual print requests by request ID, or all requests waiting to be printed on a particular printer.

For example, to move a request with a request ID of *quick-532* to a printer named *slow*, type:

```
/usr/lib/lpmove quick-532 slow
```

The print request now has a new request ID: *slow-532*.

To move all requests on a printer named *quick* to *slow*, type:

```
/usr/lib/lpmove quick slow
```

For more information on `lpmove`, see `lpsched(C)` in the *XENIX Reference Manual*.

7.7.7 Accepting and Rejecting Print Requests: `accept`

The `accept` command allows printers or classes of printers to accept print requests made with the `lp` command. After a printer has been properly configured, you can allow it to accept print requests; however, the printer will not begin printing the requests until the `enable` command is given. If you added a printer to your system using the `lpinit` command, these steps were automatically taken. For information on `enable`, see Section 7.7.8.

For example, to have print requests accepted for a class of printers named *class1*, type:

```
/usr/lib/accept class1
```

If you want to prevent requests from being routed to a printer, you can use the `reject` command. The `-r` option allows you to send users a message explaining why a printer is out of service.

For example, to prevent printing requests from being routed to a printer, *printer4*, because of repairs, type:

```
/usr/lib/reject -r "printer4 needs repair" printer4
```

A user who requests to print a file on *printer4* will receive the following message:

```
lp:can not accept requests for destination "printer4"  
--printer printer4 needs repair
```

To find out the acceptance status of printing destinations, type:

```
lpstat -a
```

For more information on **lpstat**, see the *XENIX User's Guide*. For more information on **accept/reject**, see **accept(C)** in the *XENIX Reference Manual*.

7.7.8 Enabling and Disabling Printers

The **enable** command allows **lpsched** to print files on printers. A printer can accept requests for printing after you give the **accept** command for it, but in order to print the files, you must give the **enable** command as well.

For example, to enable a printer named *daisy*, type:

```
enable daisy
```

You can disable printers with the **disable** command. The scheduler, **lpsched**, will not send printing requests to disabled printers regardless of their acceptance status. The **-r** option allows you to send a message to users explaining why a printer has been disabled.

For example, to disable a printer named *laser* because of a paper jam, type:

```
disable -r "paper jam" laser
```

Users requesting the status of *laser* with the command **lpstat -plaser** will receive the following message:

```
printer "laser" disabled since Dec 5 10:15  
paper jam
```

For more information on these two commands, see **enable(C)** and **disable(C)** in the *XENIX Reference Manual*.

7.7.9 Printer Interface Programs

Each printer on your system must have a printer interface program. This can be a shell script, C program, or any other executable program. Your XENIX system provides a set of model interface programs. See **lp(C)**. They are written as shell scripts and can be found in */usr/spool/lp/model*. You can use the programs as they are, modify them, or write your own.

If you want to write or modify a printer interface program, the following information may be helpful.

At the time **lpsched** routes a printing request to a printer *p*, the interface program for *p* is invoked in */usr/spool/lp* in the following form:

interface/P *id user title copies options file*

<i>id</i>	Identifies the request id returned by lp .
<i>user</i>	Gives the login name of the user who made the request.
<i>title</i>	Gives an optional title chosen by the user.
<i>copies</i>	Gives the number of copies requested.
<i>options</i>	Lists printer dependent options. Each option should be separated by a blank space.
<i>file</i>	Gives the full pathname of a file to be printed.

Note

With System III compatibility (default = *y* for **lpinit**), XENIX invokes ‘interface/ *_P*’.

When the interface program is started, its standard input comes from the */dev/null* file and both standard output and standard error output are directed to the printer’s device. Devices are opened for reading, as well as writing, when file modes permit. In the case where a device is a regular file, all output is appended to the end of the file.

You may format your output through interface programs in any way you choose; however, you must ensure proper **stty** modes (terminal characteristics such as baud rate and output options). In a shell script interface, this means that the printer’s device must be open for reading, that is, it must take the standard input for the **stty** command from the device.

The file */etc/default/lpd* contains a line “**BANNERS=d**” where *d* is the number of banner pages you want to print at the front of every printing request. Interface programs should examine this file and behave accordingly.

After printing is completed, the interface program should exit with a code showing that the print job was successful. Exit codes are interpreted by the

printer scheduler, **lpsched**, as follows:

Exit Code	Meaning to lpsched
0	Print job was successful.
1 to 127	Lpsched found a problem while printing this particular request, for example, too many unprintable characters. This problem will not affect future printing requests. Lpsched notifies users by mail that there was an error in printing the request.
greater than 127	These codes are reserved for internal use by lpsched . Interface programs must not exit with codes in this range.

Finally, when problems occur in printing that are likely to affect future printing requests, the printer interface program should disable printers so that print requests are not lost. When a busy printer is disabled, the interface program retains the print requests so that they are not lost.

For more information on printer interface programs, see **lpadmin(C)** in the *XENIX Reference Manual*.

7.8 Using the Dial Program

You can configure your system to dial one of two types of modems. They are:

Tandy/Radio Shack Modems
D.C. Hayes Modem

Your XENIX system is configured to dial a Hayes Modem. If you want to dial a Tandy/Radio Shack's Modem II, DC-1200, or DC-2212, then you must reconfigure the system. Instructions for setting the XENIX system back to a Hayes modem are given at the end of this document.

7.8.1 Setting XENIX for Tandy/Radio Shack Modems

1. Login as root. Type the following commands and press RETURN after each:

```
cd /usr/lib/uucp
rm dial
ln dialTRS dial
```

- Next, you must edit the file *L-devices*. XENIX uses this file to know the available dialout lines. The *L-devices* file contains a line for each dial-out line. Each dial-out line entry has 4 fields:

Field 1	Type of Connection “ACU” for a dial-out line “D-r” for a direct connection
Field 2	Line Device /dev/tty00
Field 3	Dialer Device same as Field 2
Field 4	Baud Rate Can be 300 or 1200.

Edit the *L-devices* file using any XENIX text editor. See your XENIX documentation for information on editors.

- Remove any line that begins with “ACU.” You’ll add new lines for the dial-out modems. You do not need to add any lines for the dial-in modems.
- Add new lines that refer to the appropriate serial channel and baud rate. For example, to set the dialer for tty00, use one of these lines:

```
ACU tty00    tty00    1200    for 1200 baud
ACU tty00    tty00    300     for 300  baud
```

- Connect the modem to the appropriate serial channel on your computer.

You can now follow the instructions given in your XENIX documentation for using modems.

7.8.2 Using a Hayes Modem

To set XENIX back to dialing a Hayes modem follow these steps:

- Log in as root. Type the following commands and press RETURN after each:

```
cd /usr/lib/uucp
rm dial
ln dialHAYES dial
```

- Next, you must edit the file *L-devices*. XENIX uses this file to know the available dial-out lines. The *L-devices* file contains a line for each dial-out line. Each dial-out line entry has 4 fields:

Field 1	Type of Connection “ACU” for a dial-out line “DIR” for a direct connection
Field 2	Line Device /dev/tty00
Field 3	Dialer Device Set the same as Line Device.
Field 4	Baud Rate Can be 300 or 1200.

Edit the *L-devices* file using any XENIX text editor. See your XENIX documentation for information on editors.

- Remove any line that begins with “ACU.” You’ll add new lines for the dial-out modems. You do not need to add any lines for the dial-in modems.
- Add new lines that refer to the appropriate serial channel and baud rate. For example, to set the dialer for tty00, use one of these lines:

```
ACU tty00    tty00    1200    for 1200 baud
ACU tty00    tty00    300     for 300  baud
```

If your modem operates at only 300 baud, do not add the line for 1200 baud.

- Connect the modem to the appropriate serial channel on your computer.
- Set the Hayes Modem switches as described next:

Switch	Set	Description
1	UP	This setting is preferred if your cable connects DTR (pin 20) between the modem and the computer.
	DOWN	This setting forces DTR on if DTR is not connected, such as with a 3-wire cable. With this setting, the modem may not hang up if the other end doesn’t hang up when the session is over.

2	UP	This setting is required for automatic dialing. Results are sent as English words.
3	DOWN	This setting is required for automatic dialing. Results codes are sent.
4	DOWN	This setting does not allow echoing characters in the local command mode.
5	DOWN	Modem won't answer incoming calls.
	UP	Modem answers incoming calls. This setting does not affect automatic dialing. However, UP is required if the line will be used as a dial-in line.
6	UP	This setting causes the Carrier Detect Line to reflect the real state of the carrier. This allows the computer to log a user out if the connection is broken when they dial in. Your cable must allow for Carrier Detect (pin 8 from the modem) for this to work properly.
7	UP	This setting is used for single-line telephone installations. It can differ depending on your telephone line.
8	DOWN	This setting enables command recognition. It is required for automatic dialing.

You should proceed to the section "Dial-in vs Dial-out."

7.8.3 Dial-in vs Dial-out

A serial communication line connected to a modem may be used as either a dial-in line or a dial-out line, but not both. If you plan to ever use the modem as a dial-in line, check the entry for that serial line in the */etc/tty*s file. For most modems the second character of the entry should be a '3' for 1200/300. What this means is that XENIX tries 1200 baud until a Modem Break is received. Then it tries 300 baud. There are other codes you can use instead. They are:

Code	Baud Rates
1	300
2	300/1200
3	1200/300
4	2400

For a complete list of baud rates and their codes, see **getty** (M).

To change a serial line from a dial-in line to a dial-out line, log in as root and type this command:

```
disable tty0x
```

Replace x with the appropriate serial line number. You should wait at least one minute after entering the above command before entering any other enable or disable command on any other communication line.

7.8.4 Using cu

This example illustrates using **cu** with *tty00* at 1200 baud:

```
cu -s 1200 phone-number
```

You must use **cu** with the Hayes modems.

7.8.5 Example Dial Source

The dial program is used by dial-out sites to place calls to other computers. You can create a new dial program for any dial-out site not using the Hayes Smartmodem by modifying the source program and compiling it with the C compiler **cc** that is provided with the Software Development package. See the **cc** (CP) command in the *XENIX Reference Manual Volume II* for more information. The dial source program is included on the following pages. The dial source is also in */usr/lib/uucp* along with a supporting **make** file.

D.C. Hayes is a trademark of Hayes Microcomputer Products, Inc.

```

/*
 *   Copyright (C) Microsoft Corporation, 1983
 *
 *   Simple dialer program for the Hayes "Smart" Modem 1200
 *
 *   See Hayes manual for command definitions
 *
 *   Usage:  dial ttyname telnumber speed
 *
 *   returns 0 if a connection was made
 *           -1 otherwise
 */

#include <stdio.h>
#include <signal.h>
#include <fcntl.h>
#include <sys/types.h>
#include <sys/ioctl.h>
#include <termio.h>
# include "config.h"
# include "result.h"

char *sprintf();

# define NRETRIES 3
struct termio term;
int baudrate;                      /* baud rate of modem */

main(argc,argv)
int argc;
char *argv[];
{
    FILE *fdr,*fdw;
    int  fd;
    int  ret;
    int  i;
    char buf[80];
    int  hflag = 0;

    if (argc == 5)
    {
        if (strcmp(argv[1], "-h") == 0)
        {
            hflag++;
            argc--;
            argv++;
        }
    }

```

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```
} else if (argc == 4)
    ; /* normal dialing invocation */
else
{
    fprintf(stderr, "Usage: dial devicename [number] speed\n");
    exit(-1);
}
if( (fd=open(argv[1],O_RDWR|O_NDELAY)) < 0 ) {
    fprintf(stderr, "dial: Can't open device: %s for reading.\n", argv[1]);
    exit(-1);
}
switch(atoi(argv[3])) {
    case 300:
        baudrate = B300;
        break;
    case 1200:
        baudrate = B1200;
        break;
    case 2400:
        baudrate = B2400;
        break;
    default:
        baudrate = B1200;
}
/*
 * set line for no echo and specific speed
 */
ioctl(fd, TCGETA, &term);
term.c_cflag &= ~CBAUD;
term.c_cflag |= CLOCAL|HUPCL|baudrate;
term.c_lflag &= ~ECHO;
term.c_cc[VMIN] = 1;
term.c_cc[VTIME] = 0;
ioctl(fd, TCSETA, &term);
fcntl(fd, F_SETFL, fcntl(fd, F_GETFL, 0) & ~O_NDELAY);
if( (fdr=fopen(argv[1], "r") ) == NULL ) {
    fprintf(stderr, "dial: Can't open device: %s for reading.\n", argv[1]);
    exit(-1);
}
if( (fdw=fopen(argv[1], "w") ) == NULL ) {
    fprintf(stderr, "dial: Can't open device: %s for writing.\n", argv[1]);
    exit(-1);
}
setbuf(fdw, 0); /* Want unbuffered I/O */

if (hflag)
{
    hangup(fdr, fdw);
}
```



```

        exit(0);
    }
    for (i = 0; i < NRETRIES; i++)
    {
        cmd(fdw, "Z");          /* software reset modem */
        if (result(fdr, TIMCMD) == DC_OK)
            break;
    }
    sleep(2);                  /* let the modem reset */
    for (i = 0; i < NRETRIES; i++)
    {
        cmd(fdw, "E0F1Q0V1X0");
        ret = result(fdr, TIMCMD);
        if (ret == DC_OK)
            break;
    }
    if (ret != DC_OK)
    {
        hangup(fdr, fdw);
        exit(-1);
    }

    sprintf(buf, "DT%s", argv[2]); /* dial */
    for (i = 0; i < NRETRIES; i++)
    {
        cmd(fdw, buf);
        ret = result(fdr, TIMCONN);
        if (ret == DC_CONNECT || ret == DC_NOCARR)
            break;
    }
    if (ret != DC_CONNECT)
    {
        hangup(fdr, fdw);
        exit(-1);
    }

    ioctl(fd, TCGETA, &term);
    term.c_cflag &= ~CLOCAL;
    ioctl(fd, TCSETA, &term);
    exit(0);
}

hangup(fdr, fdw)
FILE *fdr, *fdw;
{
    int i;

    for (i = 0; i < NRETRIES; i++)

```

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```
{
    cmd(fdw, "Q0");
    if (result(fdr, TIMCMD) == DC_OK)
        break;
    sleep(1);
    fprintf(fdw, "+++");
    fflush(fdw);
    if (result(fdr, TIMCMD) == DC_OK)
        break;
}
for (i = 0; i < NRETRIES; i++)
{
    cmd(fdw, "H0E0F1");
    if (result(fdr, TIMCMD) == DC_OK)
        break;
}
cmd(fdw, "Q1");
result(fdr, 2);          /* timeout expected here */
cmd(fdw, "Q1");
result(fdr, 2);          /* timeout expected here */
cmd(fdw, "Q1");
}
cmd(fd, str)
FILE *fd;
char *str;
{
# ifdef DEBUG
    printf("cmd: AT%s ...", str);
# endif DEBUG
    fprintf(fd, "AT%s\r", str);
    fflush(fd);
}
```

Chapter 8

Solving System Problems

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8.1 Introduction

This chapter explains how to solve problems that affect the operation of the system. The problems range in complexity from how to fix a nonechoing terminal to how to restore lost system files.

8.2 Restoring a Nonechoing Terminal

A nonechoing terminal is any terminal that does not display characters typed at the keyboard. This abnormal operation can occur whenever a program stops prematurely as result of an error, or when the user presses the BREAK key.

To restore the terminal to normal operation, follow these steps:

1. Press the CTRL-J key sequence. The system may display an error message. If it does, ignore the message.
2. Type:

```
stty sane
```

and press the CTRL-J key sequence. The terminal does not display what you type, so type carefully.

After pressing the CTRL-J key sequence, the terminal should be restored and you may continue your work.

8.3 Solving Lineprinter Problems

No printing can be done on the lineprinter spooling system unless the print scheduler, **lpsched**, is running. To check the status of **lpsched**, type:

```
lpstat -r
```

To restart **lpsched**, type the following on two lines:

```
rm -f /usr/spool/lp/SCHEDLOCK
/usr/lib/lpsched
```

The lock file */usr/spool/lp/SCHEDLOCK* prevents more than one version of **lpsched** from running at any one time.

Access to files and directories in */usr/spool/lp* by **lp** can be another source of spooling problems. You can check the **lpsched** log file, */usr/spool/lp/log*. This is a record of the print scheduler's activity and errors. If **lpsched** refuses to run or a printer refuses to print, check to make sure that:

- The printer is enabled; see **lp(C)** in the *XENIX Reference Manual*.
- The files and directories in */usr/spool/lp* are readable and writable by **lp**.

For more information on the lineprinter spooling system, see Section 7.7 in Chapter 7, “Using Peripheral Devices.”

8.4 Stopping a Runaway Process

A runaway process is a program that cannot be stopped from the terminal at which it was invoked. This occurs whenever an error in the program “locks up” the terminal, that is, prevents anything you type from reaching the system.

To stop a runaway process in a multi-terminal system, follow these steps:

1. Go to a terminal that is not locked up.
2. Log in as the super-user.
3. Type:

```
ps -ef
```

and press the RETURN key. The system displays all current processes and their process identification numbers (PIDs). Find the PID of the runaway program.

4. Type:

```
kill PID
```

and press the RETURN key. The *PID* is the process identification number of the runaway program. The program should stop in a few seconds. If the process does not stop, type:

```
kill -9 PID
```

and press the RETURN key.

The last step is sure to stop the process, but may leave temporary files or a nonechoing terminal. To restore the terminal to normal operation, follow the instructions in Section 8.2 in this chapter.

To stop a runaway process in a single-terminal system, you must press the reset button, or turn off the power to the system. You will need to restart the system by following the instructions in Chapter 2, “Starting and Stopping the System.”

8.5 Replacing a Forgotten Password

The XENIX operating system does not provide a way to decipher an existing password. If a user forgets a password, the system manager must change the password to a new one. To change an ordinary user password, follow the instructions in Section 3.3 in Chapter 3, “Preparing XENIX for Users.”

8.6 Removing Hidden Files

A hidden file is any file whose name begins with a period (.). You can list the hidden files in a directory by typing:

```
lc -a
```

and pressing the RETURN key.

You can remove most hidden files from a directory by typing:

```
rm .[a-z]*
```

and pressing the RETURN key. Remaining files can be removed individually with the `rm` command.

8.7 Restoring Free Space

The system displays an “out of space” message whenever the root directory has little or no space left to work. To restore system operation, you must delete one or more files from the root directory. To delete files, follow the steps outlined in Section 5.2 in Chapter 5, “Maintaining File Systems.”

8.8 Restoring Lost System Files

If a system program or data file is accidentally modified or removed from the file system, you can recover the file from the periodic backup disk with the `sysadmin` program. To restore the files, follow the instructions in Section 6.3.3 in Chapter 6, “Backing Up File Systems.”

8.9 Restoring an Inoperable System

On very rare occasions, one or more of the critical XENIX system files may be accidentally modified or removed, preventing the system from operating. In such a case, you must reinstall the XENIX system and restore user program and data files from backup disks. To reinstall the system, follow the instructions in the *XENIX Installation Guide*. To restore files from backup disks, follow the instructions in Section 6.3.3 in Chapter 6, “Backing Up File Systems.”

8.10 Recovering from a System Crash

A system crash is a sudden and dramatic disruption of system operation that stops all work on the computer. System crashes occur very rarely. They are usually the result of hardware errors or damage to the root file system which the operating system cannot correct by itself. When a system crash occurs, the system usually displays a message explaining the cause of the error, then stops. This gives the system manager the chance to recover from the crash by correcting the error (if possible) and restarting the system.

A system crash has occurred if 1) the system has displayed at the system console a message beginning with “panic:”, or 2) the system refuses to process all input (including INTERRUPT and QUIT keys) from the system console and all other terminals.

To recover from a system crash, follow these steps:

1. Use the error message(s) displayed on the system console to determine the error that caused the crash. If there is no message, skip to step 3.
2. Correct the error, if possible. A complete list of error messages and descriptions for correcting the errors is given in **messages** in the *XENIX Reference Manual*. (Even if the problem cannot be located or corrected, it is generally worthwhile to try to restart the system at least once by completing the remaining steps in this procedure.)
3. Press reset or turn off the computer and follow the steps described in Chapter 2, “Starting the System,” to restart the system.
4. If the system will not restart, or crashes each time it is started, the operating system is inoperable and must be reinstalled. Follow the procedures described in the *XENIX Installation Guide* to reinstall the system and in Chapter 6, “Backing Up File Systems,” to restore user’s files.
5. If the system cannot be started from the “Boot” disk in the distribution set for installation, the computer has a serious hardware malfunction. Contact a hardware service representative for help.

8.11 Changing XENIX Initialization

One common problem is how to adapt the system initialization to suit your system environment. This problem occurs whenever you add new devices such as terminals or disk drives to the system, and wish them to be automatically enabled or mounted whenever you start normal system operation. You can adapt system initialization by modifying the system initialization files.

The XENIX initialization files contain XENIX commands and/or data which the system reads at system startup or whenever a user logs in. The files typically mount file systems, start programs, and set home directories and terminal types. The initialization files are named */etc/rc*, *.profile*, */etc/profile*, */usr/lib/mkuser/mkuser.prof*, and */etc/motd*.

The system manager may modify these files to create any desired initial environment. The files are ordinary text files and may be modified using a text editor such as *ed* (see the *XENIX User's Guide*). Note, however, that the */etc/rc* and *.profile* files contain XENIX commands and comments, and have the command file format described in the *XENIX User's Guide*.

8.11.1 Changing the */etc/rc* File

The */etc/rc* file contains XENIX system initialization commands. The system executes the commands at system startup. The commands display a startup message, start various system daemons, and mount file systems. You can display the contents of the file with the **more** command. Type:

```
more /etc/rc
```

and press the RETURN key.

You may change the contents of the file so that the system executes any set of commands you wish. For example, if you want the system to automatically mount a new file system, simply append the appropriate **mount** command in the file. The system will execute the command on each startup.

To append a command to the file, follow these steps:

1. Log in as the super-user.
2. Invoke a text editor and specify the */etc/rc* as the file to be edited.
3. Locate the place in the file you wish to insert the command (e.g., if the command mounts a file system, insert it with other mounting commands).
4. Insert the command on a new line. Make sure you type the command correctly. The system will reject any incorrect command and all following commands when it reads the file at system startup.
5. Exit the editor.

No other changes to the file are required. Be careful not to delete any commands already in the file unless you are sure they are not needed.

8.11.2 Changing the *.profile*, */etc/profile*, and */usr/lib/mkuser/mkuser.prof* Files

The *.profile*, */etc/profile*, and */usr/lib/mkuser/mkuser.prof* files contain commands that initialize the environment for each user. The commands in these files are executed whenever the user logs in. The files usually contain commands that set and export various system variables (e.g., *TERM*, *PATH*, *MAIL*). These variables give the system information such as what terminal type is being used, where to look for programs the user runs, where to look for the user's mailbox, what keys to expect for the "kill" and "backspace" functions, and so on (see the *XENIX User's Guide*).

There is one *.profile* file for each user account on the system. The files are placed in the user's home directory when the account is created. Ordinary users may modify their own *.profile* files, or allow the system manager to make modifications. In either case, the file can be edited the same way as the */etc/rc* file, using a text editor. Commands can be added or removed as desired.

8.11.3 Changing the */etc/motd* File

The message-of-the-day file, */etc/motd*, contains the greeting displayed whenever a user logs in. Initially, this file contains the name and version number of the XENIX system. It can be modified to include such messages as a reminder to clean up directories, a notice of the next periodic backup, and so on.

The */etc/motd* file is an ordinary text file, so you can change the message by editing the file with a text editor. One common change is to include a reminder to delete unused files in order to preserve disk space. In general, you should limit the size of the file to include no more than a screenful of information.